SURFACE APPROXIMATION

KRISTOPHER WEHAGE*

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ABSTRACT

This document discusses techniques to represent a terrain height field using a rectangular array of sampled elevation data points $z(x_i, y_i)$. Bilinear, bicubic and Catmul-Rom interpolation methods are discussed.

* *Robotics Technologist, Jet Propulsion Laboratory*

1 INTRODUCTION

In this document, the name *terrain map* is used to describe a rectangular array of sampled elevation data points $z_i = f(x_i, y_i)$. Terrain maps are also sometimes referred to as elevation maps, displacement maps, or height fields. This document discusses techniques to compute interpolated positions, surface normals and penetration depths of terrain maps using bilinear, bicubic and Catmul-Rom interpolation.

2 BILINEAR INTERPOLATION

Bilinear interpolation is an extension of linear interpolation. In 2D bilinear interpolation, interpolation is performed along a first axis, followed by a second axis. Note that contrary to whats its name would suggest, the surface obtained from bilinear interpolation is *not* linear; rather it is a quadratic function obtained from the product of two linear functions. Compared to other techniques, bilinear interpolation has the advantage of being relatively fast to compute. Additionally, the interpolated surface obtained from bilinear interpolation passes exactly through the control points, which may be desirable for some applications. A disadvantage to bilinear interpolation is that the resulting surface is only C_1 continuous between adjacent surface patches (i.e. continuous in position), but not C_2 or C_3 continuous (i.e. continuous in slope or curvature).

2.1 Interpolated Position

Consider a surface patch represented with the four control points $f_{11} = f(x_1, y_1)$, $f_{12} = f(x_1, y_2)$, $f_{21} = f(x_2, y_1)$, and $f_{22} = f(x_2, y_2)$. To obtain the interpolated value at position (x, y), first perform interpolation along the x-direction as

$$f(x, y_1) \approx \frac{x_2 - x}{x_2 - x_1} f_{11} + \frac{x - x_1}{x_2 - x_1} f_{21} \tag{1}$$

$$f(x, y_2) \approx \frac{x_2 - x}{x_2 - x_1} f_{12} + \frac{x - x_1}{x_2 - x_1} f_{22} \tag{2}$$

Interpolating Eqs. (1)-(2) along the y-direction yields

$$\begin{aligned} f(x,y) \approx & \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2) \\ = & \frac{y_2 - y}{y_2 - y_1} \left(\frac{x_2 - x}{x_2 - x_1} f_{11} + \frac{x - x_1}{x_2 - x_1} f_{21} \right) + \frac{y - y_1}{y_2 - y_1} \left(\frac{x_2 - x}{x_2 - x_1} f_{12} + \frac{x - x_1}{x_2 - x_1} f_{22} \right) \\ = & \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left(f_{11}(x_2 - x)(y_2 - y) + f_{21}(x - x_1)(y_2 - y) + \right. \\ & \left. f_{12}(x_2 - x)(y - y_1) + f_{22}(x - x_1)(y - y_1) \right) \\ = & \frac{1}{(x_2 - x_1)(y_2 - y_1)} \begin{Bmatrix} x_2 - x & x - x_1 \end{Bmatrix} \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} \begin{Bmatrix} y_2 - y \\ y - y_1 \end{Bmatrix} \end{aligned} \tag{3}$$

Substituting $u = \frac{x - x_1}{x_2 - x_1}$, $v = \frac{y - y_1}{y_2 - y_1}$ into Eq. (3) effectively normalizes the interpolation to a range from 0 to 1 along the x and y directions, yielding a simpler expression as

$$f(u, v) = \begin{Bmatrix} 1 - u & u \end{Bmatrix} \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} \begin{Bmatrix} 1 - v \\ v \end{Bmatrix} \tag{4}$$

Expanding Eq. (4) yields

$$\begin{aligned} f(u,v) &= f_{11}(1-u)(1-v) + f_{21}u(1-v) + f_{12}(1-u)v + f_{22}uv \\ &= (f_{11} - f_{11}u)(1-v) + f_{21}u - f_{21}uv + f_{12}v - f_{12}uv + f_{22}uv \\ &= f_{11} - f_{11}u - f_{11}v + f_{11}uv + f_{21}u - f_{21}uv + f_{12}v - f_{12}uv + f_{22}uv \\ &= f_{11} + (-f_{11} + f_{21})u + (-f_{11} + f_{12})v + (f_{11} - f_{21} - f_{12} + f_{22})uv \\ &= a_{11} + a_{21}u + a_{12}v + a_{22}uv \end{aligned} \tag{5}$$

where

$$\begin{aligned} a_{11} &= f_{11} \\ a_{21} &= f_{21} - f_{11} \\ a_{12} &= f_{12} - f_{11} \\ a_{22} &= f_{11} - f_{21} - f_{12} + f_{22} \end{aligned} \tag{6}$$

2.2 Interpolated Normal

The normal can be obtained from the tangent vectors computed at (x, y) or alternatively using the normalized coordinates (u, v). Let $\underline{z}$ represent a point on the interpolated surface.

$$\underline{z} = \begin{Bmatrix} x \\ y \\ f(x,y) \end{Bmatrix} = \begin{Bmatrix} u(x_2 - x_1) + x_1 \\ v(y_2 - y_1) + y_1 \\ f(u,v) \end{Bmatrix} \tag{7}$$

The tangent vector in the x-direction can be obtained as

$$\underline{t}_x = \frac{\partial \underline{z}}{\partial x} = \frac{\partial u}{\partial x}\frac{\partial \underline{z}}{\partial u} = \frac{1}{x_2 - x_1}\begin{Bmatrix} x_2 - x_1 \\ 0 \\ a_{21} + a_{22}v \end{Bmatrix} = \begin{Bmatrix} 1 \\ 0 \\ \frac{a_{21} + a_{22}v}{x_2 - x_1} \end{Bmatrix} \tag{8}$$

Similarly the tangent vector in the y-direction can be obtained as

$$\underline{t}_y = \frac{\partial \underline{z}}{\partial y} = \frac{\partial v}{\partial y}\frac{\partial \underline{z}}{\partial v} = \frac{1}{y_2 - y_1}\begin{Bmatrix} 0 \\ y_2 - y_1 \\ a_{12} + a_{22}u \end{Bmatrix} = \begin{Bmatrix} 0 \\ 1 \\ \frac{a_{12} + a_{22}u}{y_2 - y_1} \end{Bmatrix} \tag{9}$$

ISO CONVENTION In the ISO convention, the surface normal $\underline{n}$ points in the positive z direction and is obtained as

$$\begin{aligned} \underline{n} = \tilde{\underline{t}}_x \underline{t}_y &= \begin{bmatrix} 0 & -t_{x(2)} & t_{x(1)} \\ t_{x(2)} & 0 & -t_{x(0)} \\ -t_{x(1)} & t_{x(0)} & 0 \end{bmatrix} \begin{Bmatrix} t_{y(0)} \\ t_{y(1)} \\ t_{y(2)} \end{Bmatrix} \\ &= \begin{Bmatrix} t_{x(1)}t_{y(2)} - t_{x(2)}t_{y(1)} \\ t_{x(2)}t_{y(0)} - t_{x(0)}t_{y(2)} \\ t_{x(0)}t_{y(1)} - t_{x(1)}t_{y(0)} \end{Bmatrix} = \begin{Bmatrix} -\frac{a_{21} + a_{22}v}{x_2 - x_1} \\ -\frac{a_{12} + a_{22}u}{y_2 - y_1} \\ 1 \end{Bmatrix} \end{aligned} \tag{10}$$

The normal vector is reported as a normalized quantity by dividing by its length.

$$\underline{n}_{ISO} = \frac{\underline{n}}{||\underline{n}||} \tag{11}$$

SAE CONVENTION In the SAE convention, the surface normal $\underline{n}$ points in the negative z direction and is obtained as

$$\begin{aligned}\underline{n} = \underline{\tilde{t}}_y \underline{t}_x &= \begin{bmatrix} 0 & -t_{y(2)} & t_{y(1)} \\ t_{y(2)} & 0 & -t_{y(0)} \\ -t_{y(1)} & t_{y(0)} & 0 \end{bmatrix} \begin{Bmatrix} t_{x(0)} \\ t_{x(1)} \\ t_{x(2)} \end{Bmatrix} \\ &= \begin{Bmatrix} t_{y(1)}t_{x(2)} - t_{y(2)}t_{x(1)} \\ t_{y(2)}t_{x(0)} - t_{y(0)}t_{x(2)} \\ t_{y(0)}t_{x(1)} - t_{y(1)}t_{x(0)} \end{Bmatrix} = \begin{Bmatrix} \frac{a_{21}+a_{22}v}{x_2-x_1} \\ \frac{a_{12}+a_{22}u}{y_2-y_1} \\ -1 \end{Bmatrix}\end{aligned} \tag{12}$$

Finally the normal vector is normalized for reporting purposes as

$$\underline{n}_{SAE} = \frac{\underline{n}}{\|\underline{n}\|} \tag{13}$$

3 BICUBIC INTERPOLATION

4 CATMUL-ROM INTERPOLATION

REFERENCES